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RESEARCH ARTICLE

A Study on Adversarial Sample Resistance and Defense Mechanism for Multimodal Learning-Based Phishing Website Detection

PHAN THE DUY^{ID}, VO QUANG MINH, BUI TAN HAI DANG^{ID}, NGO DUC HOANG SON^{ID},
NGUYEN HUU QUYEN^{ID}, AND VAN-HAU PHAM^{ID}

Information Security Laboratory, University of Information Technology, Vietnam National University Ho Chi Minh City (UIT-VNU-HCM), Ho Chi Minh City 720400, Vietnam

Corresponding author: Van-Hau Pham (haupv@uit.edu.vn)

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ABSTRACT Recent advancements in Artificial Intelligence (AI) have greatly impacted cybersecurity, particularly in detecting phishing websites. Traditional methods struggle to address evolving vulnerabilities, but research shows that Machine Learning (ML), Ensemble Learning (EL), and Deep Learning (DL) are effective in developing defenses. However, these methods face challenges with adversarial examples (AEs). The multimodal model (MM) is a promising solution, yet there is a significant lack of research using multimodal techniques specifically for phishing website detection (PWD) against adversarial websites. To tackle this challenge, this paper assesses 15 learning-based models, particularly multimodal ones, for phishing and adversarial detection, aiming to enhance their defense capabilities. Due to the scarcity of adversarial websites, training and testing models are limited. Therefore, this study proposes an innovative attack framework, AWG - Adversarial Website Generation that employs Generative Adversarial Networks (GAN) and transfer-based black box attacks to create AEs. This framework closely mirrors real-world attack scenarios, ensuring high effectiveness and realism. Finally, we present defense strategies with straightforward implementation and high effectiveness to enhance the resistance of models. The models underwent training and testing on a dataset collected from reputable sources such as OpenPhish, PhishTank, Phishing Database, and Alexa. This approach was chosen to ensure the dataset's diversity and relevance to reflect real-world conditions. Experimental results highlight that the Generator's effectiveness is demonstrated by a domain structure generation rate exceeding 90%. Moreover, AEs generated by this Generator effectively bypass most state-of-the-art ML, DL, and EL models with an evasion rate of up to 88%. Notably, the Support Vector Machine (SVM) model is the most vulnerable, with a detection rate of only 10.02%. On the other hand, the MM Shark-Eyes demonstrates outstanding resistance against AEs, with a detection rate of up to 99%. Upon applying our defense strategy, the resistance of models is significantly boosted, with all detection rates surpassing 90%. These findings underscore the robustness of our methods and pave the way for further exploration into advanced attack and defense strategies in the context of phishing and adversarial website detection.

INDEX TERMS Multimodal, deep learning, machine learning, ensemble learning, phishing website detection, phishing domain, adversarial attacks, transferability of adversarial samples.

I. INTRODUCTION

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Phishing is a typical kind of cybercrime [1] by employs both social engineering and technical subterfuge to steal

the consumer's personal identity data and financial account credentials [2], [3]. Phishing attackers typically create fraudulent websites with domains, URLs, and appearances that closely resemble legitimate websites to deceive users and obtain their sensitive data. According to the latest report of the Anti Phishing Working Group (APWG) [2] in Q4 2023, APWG has logged more than 5 million phishing attacks. Since the beginning of 2019, the number of phishing attacks has grown by more than 150% per year. The rapid increase in the number of phishing websites is facilitated by the emergence of phishing toolkits, which enable attackers to quickly create phishing websites while ensuring necessary functionalities and increasing website quality. This poses challenges for differentiation based on human vision [4].

In the realm of cybersecurity, various techniques have emerged to detect phishing attacks [5], [6], [7]. Typically, the blocklist-based method [4], [8], [9] is the easiest to implement and has gained widespread adoption. However, these blocklists are updated periodically, which may result in delayed detection of unknown malicious phishing websites [10]. In light of this limitation, ML techniques have been introduced to detect phishing websites effectively [11]. Based on previous studies [12], there are various approaches to applying AI in phishing website detection, these include designing feature extraction frameworks suitable for web applications for ML models so that the models can operate stably, efficiently, and resource-light [13], building DL model architectures to leverage automatic feature extraction [14], or employing EL to leverage the individual strengths of different model groups [15], [16]. Among these numerous studies, the majority focus on URL-based approaches for detecting phishing websites [17], [18], while others rely on domain-based approaches [19], [20]. Some studies also concentrate on the external appearance of websites or rely on third-party sources to gather information [15]. All studies aim to provide solutions that meet the demands of real-world deployment, are highly efficient and resource-saving, require minimal effort for preparation and deployment, and effectively utilize comprehensive datasets.

However, despite their high reliability, it is widely acknowledged that ML/DL methods are susceptible to AEs [21]. Adding AEs with perturbations can cause ML-based detectors to make incorrect decisions. For instance, the ML/DL methods focusing on early URL appearance for phishing website detection continue to be extensively researched and applied due to their effectiveness. However, despite these shining points, URL-based models are susceptible to evasion techniques. Sabir et al. research [22] has highlighted the weakness and vulnerability of URL-focused ML models against AEs. In the studies, SpacePhish [23] and Multi-SpacePhish [24] by the authors Apruzzese, it was demonstrated that AEs could heavily impact state-of-the-art ML-PWD models. This was shown through the implementation of 12 different adversarial attack types. This highlights the challenge faced by AI research in identifying phishing websites.

To address adversarial attacks, MMs present a promising solution. They focus on various entities to extract features. For example, the study by Revue [25], which concentrates on URLs along with JavaScript source and website content, or the Shark-eyes model [26], which focuses on HTML tag groups and domains for feature extraction. Yuan et al. [27] have proposed a multimodal approach that combines image feature extraction and structural feature extraction representation to detect malicious URLs. These studies have achieved positive results and show promising potential for robustness against phishing websites. However, there is a lack of research on the resilience of MMs against adversarial attacks in the context of Pwd. For instance, although SpacePhish [23] and Multi-SpacePhish [24] represent the first statistically validated assessments of state-of-the-art ML-PWD against 12 evasion attacks, they still lack validated assessments involving modern DL, EL, and MM models. Furthermore, there is a limited number of publicly available AEs, especially in the context of phishing websites. Therefore, to improve the quality of learning-based models, we propose an attack framework to generate effective AEs and a defense strategy to enhance the robustness of these models.

In this paper, we aim to build an attack framework capable of generating diverse adversarial patterns and samples that can bypass various ML models and be applied in real-world scenarios. Therefore, we employ three main techniques to generate AEs: using GAN models, black-box attacks, and transfer-based attacks. Particularly, we enhance the attack strategy known as PWDGAN [28] and integrate it with the malicious domain generation algorithm called Charbot [29]. The generated AEs will be utilized to attack both state-of-the-art unimodal and multimodal models.

In our endeavor to enhance the efficiency of fraudulent website detection systems amidst the increasingly adversarial attack patterns. Our study contributes significantly in the following ways:

- We introduce AWG, an adversarial attack framework utilizing Wasserstein Generative Adversarial Networks (WGAN) to generate AEs for fooling phishing website detectors. This model focuses on simulating a realistic attack phase while ensuring high effectiveness, thereby challenging the phishing website detection systems based on blocklist databases and AI-based approaches.
- By assessing the resilience of contemporary architectural models, including both unimodal and multimodal, against adversarial attacks in the scam website detection context, we provide a comprehensive insight into these models' defensive capabilities against new and complex attack patterns.
- Moreover, this work also explores the transferability of adversarial examples across non-targeted detection models, illuminating how these attack patterns can affect various systems without requiring detailed information about the targeted model.
- Building upon our findings, we propose an effective defensive mechanism to augment the discriminatory

capabilities against adversarial patterns using adversarial training. This method not only significantly improves the detection capabilities of current models but also paves the way for developing more advanced defense strategies in the future.

The remaining sections of this article are constructed as follows. Section II introduced some related works in phishing detection. In Section III, we present the fundamental technology driving our research. Next, the methodology is discussed in Section IV. Section V describes the experiments and analysis. Finally, we conclude the paper in Section VII.

Our source code is available in a GitHub link: <https://github.com/dangbuiUIT/AWG-Framework>.

II. RELATED WORK

A. PHISHING WEBSITE DETECTION

Methods for detecting phishing on the Internet are increasingly being researched by cybersecurity experts [30]. Among them, the method of fraud through websites is being utilized more and more by attackers. Traditional methods of detecting phishing websites often rely on simple techniques, such as based on the similarity of the website interface or using black/white lists that various studies have researched [7], [9], [31], [32]. These methods often face limitations in terms of accuracy, the ability to identify new phishing websites, and challenges in detecting embedded objects. Furthermore, the techniques of attackers are becoming increasingly sophisticated and complex. They can rapidly create a series of phishing websites in a short period, mimicking the interface and functionality of legitimate websites. This adds to the difficulty of detecting and preventing fraudulent online activities [33]. Therefore, there is a high demand for new phishing detection approaches to address these challenges.

ML and DL technologies have been applied to save time as well as enhance the capability of detecting fraudulent websites [34], [35], [36], [37]. For instance, in the research [36], they utilized a combination of ML and a blocklist method to improve the predictive ability for identifying deceptive websites. This method achieved an impressive F1-score of 84%, demonstrating a significant improvement when applying ML. In addition, studies integrating DL into fraud detection have also received high praise. In a conducted study [34], the authors developed DL models, including CNN, LSTM, and CNN-LSTM, for detecting counterfeit websites. The results indicated that the accuracy of these models exceeded 96%, with CNN outperforming impressively at an accuracy rate of 99.2%. In the research [31], a DL model using VAE-DNN is proposed to reduce the dimensionality of one-hot encoded input data, enhancing the efficiency and effectiveness of feature learning from URL data. Furthermore, the CNN model proposed in the study [18] employs 1D convolutions to extract features from URLs. Both studies have achieved significant effectiveness in phishing detection. However, they solely focus on URL structures to detect fraudulent websites. This leads to attackers being able to easily bypass detection systems by conducting adversarial attacks [21], [38].

To address the issues arising from previous solutions, multimodal approaches have emerged as a recent trend in Artificial Intelligence (AI) and are gaining attention from researchers due to their promising performance. Thanks to multiple perspectives from various models, MM offers the potential for high accuracy [39]. For example, Zhang et al. [40] provide MultiPhish, the first study to combine neural networks and multi-modal characteristics for phishing detection. Moreover, Feng et al. [41] trained a hybrid DL model with features extracted from URLs, HTML content, and the DOM tree. By combining BiLSTM and CNN models sequentially, the proposed model by the authors achieves an accuracy rate of 99.02% with a False Positive Rate (FPR) as low as 0.25%. Yang et al. [42] introduced a Multidimensional Feature Phishing Detection (MLPD) driven by DL to protect against phishing websites. The outcomes indicate that MLPD not only guarantees a high accuracy of 98.99% but also attains a minimal FPR of only 0.59%, all while sustaining excellent real-time performance. Furthermore, in a parallel configuration, the authors [27] explored the effectiveness of the MM in the context of identifying malicious URLs. According to the experimental findings, the approach demonstrated an accuracy of 99.78%, outperforming other methods. Another notable example is the MM proposed by Zhang et al. [43]. Their approach relies on hierarchical fusion features extracted from various aspects, including domain name, text, structure, and logo image of the webpage. After experimenting with various scenarios, the results demonstrate that their solution is highly effective in detecting phishing domain names compared to others.

As indicated in prior research, various implementation structures for multimodal architecture have been investigated. These strategies can be primarily classified into two groups: parallel and sequential. Both approaches provide high accuracy, low FPR, and effective real-time detection, thereby minimizing the risk of overlooking deceptive websites by utilizing features from different aspects of a website. Nonetheless, research on the adversarial robustness capabilities of MM is still relatively limited.

In this paper, we built and evaluated various AI models, including ML, DL, EL, and MM approaches. Our focus was on detecting phishing domains and websites, and we used a carefully curated dataset for evaluation. We aimed to assess these models in detecting both phishing and adversarial websites, addressing the scarcity of research on multimodal approaches in these fields.

B. ADVERSARIAL EXAMPLES ATTACKS

Recently, AEs have attracted significant attention owing to their capacity to exploit flaws in numerous existing ML/DL classifiers [44]. Previous studies [21], [38] have discovered that ML/DL models are often susceptible to attacks by injecting adversarial noise into the input, leading to unreliable classification results. For this reason, detection systems are becoming more vulnerable as AEs are increasingly utilized to deceive ML/DL models. An AE attack can be

categorized into three types: white-box, gray-box, or black-box attacks, depending on the information of the attacker about the target model [45], [46]. In the white-box setting, the attacker fully knows the target model, including its architecture, parameters, and gradients of the loss function concerning the input. In the computer vision domain, the first AE attack (FGSM - Fast Gradient Sign Method) leveraged gradient information for image classification was introduced by Goodfellow [47]. However, this type of attack is rarely used in practice because information about the target model is carefully concealed, posing obstacles to the attack process. Therefore, there have been some studies on gray-box [48] and black-box AE attacks [49]. In a gray-box attack, the attacker has partial knowledge about the target model, while in a black-box attack, the attacker has absolutely no information about the target model.

Recent advanced studies demonstrate the utilization of GANs as an innovative approach to generate samples in black-box attack settings. In detail, in the research conducted by Hu and Tan [50], MalGAN emerges as a pioneering contribution in the domain of black-box AE attacks aimed at Android malware detection. This approach utilizes a substitute model as a discriminator, and the generator is trained to craft AEs capable of evading the target black-box model. Besides that, in the social engineering attack, GANs can also be utilized to generate phishing URLs [28] with the intent of deceiving users into divulging sensitive information. These malicious URLs mimic legitimate websites or services, tricking users into believing they are interacting with a trusted source. Once the user enters their confidential data, such as login credentials or financial information, the attackers can capture and exploit this sensitive data for nefarious purposes, potentially leading to identity theft, financial fraud, or other harmful consequences for the targeted individuals. Therefore, developing robust defense mechanisms and user awareness campaigns becomes crucial to combat the ever-evolving threat of GAN-generated phishing URLs.

In light of rapidly evolving phishing techniques and AEs generation methods, models must be trained using datasets containing adversarial attack patterns. Thus, we proposed AWG utilizing WGAN and transfer-based black-box attacks to generate adversarial websites. This is the innovative feature of our study. By leveraging this technique, our model is closely mimicking real-world scenarios. These samples will undergo penetration testing against models to assess their defensive capabilities. Ultimately, we leverage these adversarial examples to enhance the resistance of learning-based models through our defense mechanism.

C. DEFENSE AGAINST ADVERSARIAL ATTACKS

Since AE attacks have been used widely in practice, many research efforts have been attempted to mitigate these consequences [51], [52], [53], [54]. There are many defense strategies in adversarial robustness. In the survey [55], the authors identified three main defense solutions: Model optimization, Data Optimization, and Additional network.

With the first solution, defenders attempt to preprocess the input to filter out perturbations or add new layers to the original model to remove noise from the samples during feature extraction. With the second solution, defenders seek to improve the input in various ways, such as mixing AEs into the dataset, reducing the dimension of the input feature vector, or altering the structure of the input. The final solution involves methods such as adding an adversarial pattern detection layer for filtering, as well as providing additional information to the original model for decision-making. This adversarial pattern detection layer needs to be trained with its dataset.

Besides, Zhang et al. [56] introduced the TIKI-TAKA framework for evaluating DL model robustness against adversarial attacks and implementing proposed defense mechanisms to enhance resistance. The defensive strategies encompass model voting ensembling, ensembling adversarial training, and query detection. Remarkably, these methods achieved nearly 100% detection of malicious traffic by intrusion detection systems, showcasing the efficacy of their approaches. This success provides a solid groundwork for future research to refine defensive techniques.

To enhance the adversarial robustness of our models, this research employs the Data Optimization technique. The techniques will be described in detail in Section IV. Based on previous research, it is evident that the retraining technique continues to be a simple, feasible, and effective method. By incorporating selected adversarial samples into the retraining process, our defense strategy demonstrates significant effectiveness, as detailed in Section V. Consequently, the models exhibit enhanced defensive capabilities against both phishing and adversarial websites.

D. MOTIVATION

Acknowledging the strengths and limitations of prior research, this study aims to incorporate their ideas and propose enhancement solutions. Details are provided in the Table 1.

Thanks to the pioneering research of these authors [23], [24], [28], [40], [57], they have shed light on the path for our advanced studies. To address the weaknesses highlighted in Table 1, we conduct experiments to evaluate the robustness of MMs against AEs. Next, we implement adversarial attacks using AWG Framework across 15 learning-based models, including ML, DL, EL, and MMs. We enhance the GAN model architecture by incorporating variants such as WGAN and techniques like transfer-based attacks, building upon the work of [28]. Furthermore, we employ Data Optimization techniques [55] to carefully select AEs during the retraining process, addressing the challenges identified in [57].

III. BACKGROUND

Based on the previous Spacephish study by Apruzzese et al. [23], the preprocessing space is the space where attackers most frequently deploy attacks and also the least costly space for the attack and attackers lack any knowledge

TABLE 1. Summary of previous works compared with our study.

Authors	Category	Contribution Summary	Limitation
Zhang et al. [40]	MultiModal Learning	Perform the first study to use multimodal features with neural networks for phishing detection.	Lack of experience with AEs
Bac et al. [28]	Adversarial Attack	Use GAN to create adversarial malicious URL for bypassing Phishing Website Detector.	No survey on DL, EL, and MM models. The metrics of the Generator can still be improved.
Apruzzese et al. [23], [24]		Conduct the first statistically validated assessment of state-of-the-art ML-PWD against 12 evasion attacks.	Lack of assessment of DL, EL and MM models.
Madry et al. [57]	Resistant to Adversarial Attacks	Propose defense methods against AEs.	Backfire if overused, as excessive retraining with many samples may degrade performance.
Our research	MultiModal Learning Adversarial Attack Defense against Adversarial	Investigate the resilience of learning-based models, particularly multimodal models against Adversarial Website. Propose an attack framework using WGAN and defense strategy to enhance the robustness of AI models.	The AWG framework should be enhanced to develop AEs with increased evasion rates. Memory and runtime optimizations are needed for the MMs to ensure better performance.

about phishing website detection models; hence, they usually deploy black-box attack methods. In this space, there are numerous ways of manipulation, ranging from adding noise to images on the website, changing URLs, altering source code, and more, all of which attackers can choose to deploy on their websites. Attackers alter various characteristics of the websites they own in multiple directions to disrupt the preprocessing process of phishing website detection models to deceive the detection models into misclassifying their phishing websites, which can then be deployed on the internet to reach potential victims.

A. MULTIMODAL LEARNING

The multimodal learning (MML) approach represents a significant advancement over monomodal (unimodal) learning by integrating data from multiple sources or modalities to improve learning accuracy and efficiency [58]. Unlike monomodal learning, which relies on a single type of data input—such as text, images, or sound—multimodal learning combines these diverse inputs to capture a more comprehensive understanding of the information being processed. This integration allows for more robust models that can handle complex, real-world scenarios more effectively [59]. There are three main types of fusion in multimodal learning: early fusion, where data from different modalities are combined before being fed into the learning model; late fusion, where predictions from separate models for each modality are combined; and hybrid fusion, which incorporates elements of both early and late fusion, optimizing the strengths of each to improve overall performance. Early fusion focuses on integrating raw data, late fusion emphasizes combining decisions from multiple models, and hybrid fusion leverages both raw data and decision-level information for a comprehensive approach. The architecture of multimodal-based learning models is depicted in Figure 1.

B. MULTIMODAL APPROACH FOR PHISHING DETECTION

Multimodal models in Phishing Website Detection (PWD) have attracted attention for their capability to integrate information from diverse sources and combine various advanced data processing techniques, resulting in high effectiveness in phishing website detection and strong robustness against

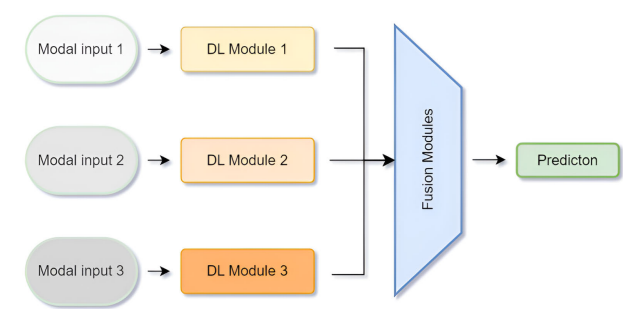


FIGURE 1. The overview of multimodal-based learning architecture.

noise. The details are represented in Figure 2, multimodal approach involves building different branches, each focusing on an entity along with a processing method to visualize the data prepared for input into machine learning algorithms. Finally, the results will be fused in various ways, such as concatenation and input into a recognition network, or voting, averaging if the branches are designed to make decisions rather than extract features.

According to statistics from previous studies, in Table 2, multimodal models typically combine feature extraction from two or more entities and employ three or more data processing methods for the collected information. The entities of interest for feature extraction include URLs, domains, content, HTML tags, Java scripts, and website appearance. In the Multiphish [40] study, a multimodal phishing website detection system operates across various languages with high performance, focusing on gathering data from two entities: the domain and favicon of the website have been introduced. The data processing of Multiphish involves combining techniques with a variation autoencoder (VAE) to enhance stability and effectiveness. The conversion of collected data into image format is also widely applied in multimodal approaches. Combining information obtained from image format data with information derived from analyzing the textual structure of the data will provide a more comprehensive and holistic view of the website. Alseadi et al. [60]. proposed a multimodal architecture that leverages both image and textual structure factors. Features extracted from URLs and DNS are processed in image and text forms, resulting in high effectiveness in phishing website detection.

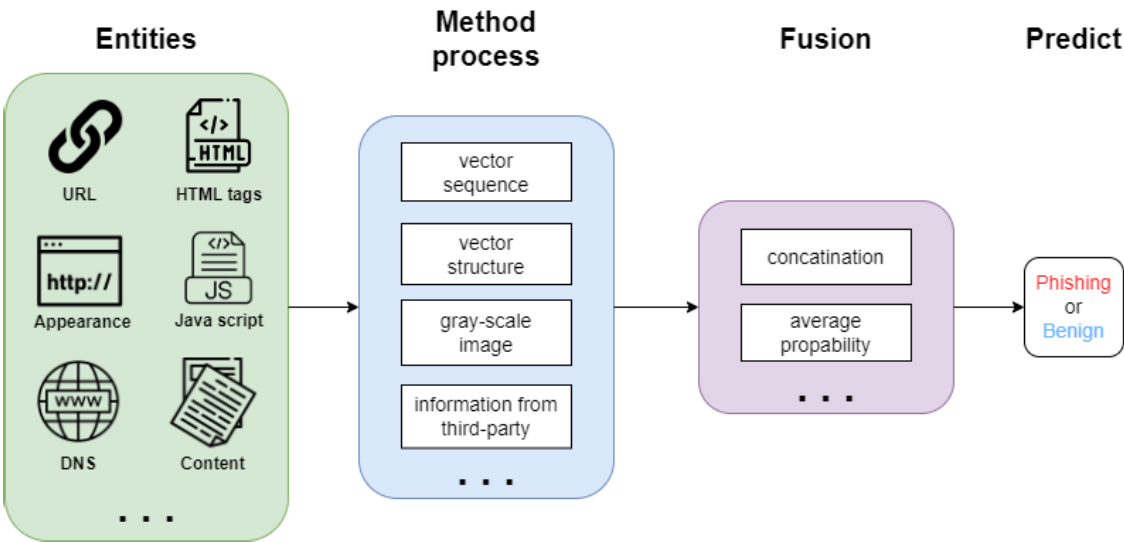


FIGURE 2. The overview of multimodal learning approach for phishing website detection.

TABLE 2. Previous research on multimodal analysis for phishing website detection.

Model	Entities	Method	Fusion	Description
Multiphish [40]	Domain Favicon	Vectorization VAE URL Feature	Concatenation (Early Fusion)	The domain and favicon are processed using a VAE to extract feature vectors. Subsequently, these two vectors are concatenated with the vector extracted from the URL to form the input data for the classification network
Shark-Eyes [26]	Domain HTML Tags	Vectorization Transform Grayscale	Concatenation (Early Fusion)	The implementation involves three branches: Branch 1 extracts grayscale image features of the domain, Branch 2 extracts syntactic structure features of the domain, and Branch 3 extracts structural features of the HTML tags. All extracted data is concatenated to create input for the classification network.
Alseadi-multimodal [60]	URL DNS	Transform Grayscale Image TF/IDF	Concatenation (Early Fusion)	The URL and DNS entities are feature-extracted in both grayscale image and text formats to capture characteristics related to appearance and sequential structure of words. The resulting data vectors are then concatenated and fed into the classification network layer.
Multi-Modal DL [25]	URL Content Javascript	Vectorization Token and AlBert Sequence	Average Propability of Branches (Late Fusion)	The implementation involves three branches: Branch 1 focuses on the syntactic structure of the URL, Branch 2 utilizes an AST parser to generate a syntax tree and extracts sequence features of JavaScript. Branch 3 deploys the AlBert natural language model to extract content features. Observations are made at each branch, and the final result is calculated based on average.

In our previous research, Shark-Eyes [26], the domain was processed in both text and image forms, combined with the analysis of the HTML tags structure, resulting in a reliable and robust multimodal system for phishing website detection. In phishing website detection, multimodal approaches hold great potential for further research. The variation in combining different attribute groups along with various data processing methods may lead to a new multimodal system that achieves high reliability and effectiveness without requiring excessive resources for real-world deployment.

IV. METHODOLOGY
A. THREAT MODEL

In our context, the threat model is defined based on four main criteria as outlined in the research by Biggio and Roli [61]. We refer to the previous Spacephish study by Apruzzese et al. [23] to build a detailed threat model in the phishing website detection field.

- **Goal:** alter various characteristics of attacker’s websites in multiple directions to disrupt the preprocessing process of the phishing website detection models to

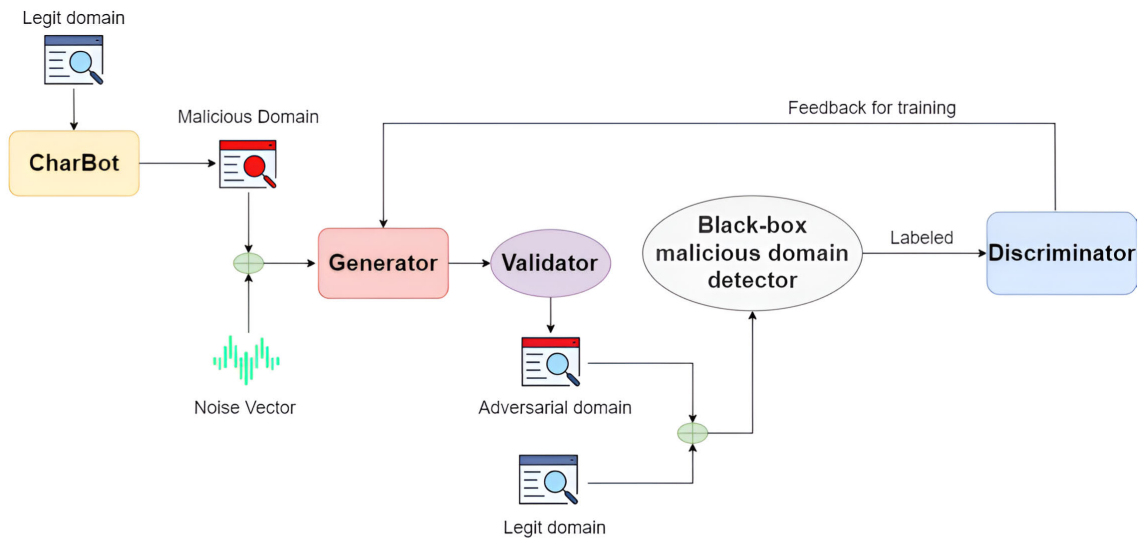


FIGURE 3. The architecture of AWG - a WGAN-based adversarial sample generation framework against black-box phishing detector.

deceive the detection models into misclassifying their phishing websites, which can then be deployed on the internet to reach internet users, who are potential victims.

- **Knowledge:** In a black-box attack scenario, the attacker has no knowledge about the victim models. They know nothing about the ML/DL model, its training data, and its underlying ML/DL algorithm.
- **Capability:** The adversary cannot access the victim model. This prevents them from inspecting its output for a given input or altering information such as hyperparameters of the victim models. This means that attackers can only manipulate their adversarial examples to find ways to generate new version samples that can bypass the victim models. There are numerous ways of manipulation, ranging from adding noise to images on the website, changing characters or words in the URL, altering source code, and more, all of which attackers can choose to deploy on their websites to make a new version that can bypass the phishing detector model.
- **Strategy:** Based on the previous Spacephish study by Apruzzese et al. [23], the preprocessing space is the space where attackers most frequently deploy attacks and also the least costly space for the attack. In this space, attackers try to mutate their website to noise the preprocessing process input data of the models. ML/DL models are often susceptible to adversarial attacks. Attackers exploit the transferability of high-quality adversarial samples to target learning-based models. They attempt to gather high-quality adversarial samples that can bypass the model **B** in order to conduct a transfer attack to target model **A** and get success.

B. THE ARCHITECTURE OF AWG MODEL

As shown in Figure 3, the architecture of our AWG framework includes the Charbot, the WGAN model with Generator, Discriminator, and the Black-box detector (*BB*). *BB* serves as a domain classifier, as in real-life scenarios, the exact ML algorithm implemented within is unknown. The Charbot algorithm generates malicious domains from benign domains, which then serve as the input for the Generator. The Generator produces new malicious domain examples, while the Discriminator emulates the *BB*, classifying examples as either legitimate or malicious, and providing information for training the Generator. The Validator's role is to verify that the domains generated by the Generator maintain the fundamental structure of a domain. We assume that the attacker does not know the *BB*. Based on the feedback from the Discriminator, the Generator gradually improves itself. After the training process, the Generator gains the capability to generate adversarial examples that can deceive the *BB*, making it challenging for the *BB* to distinguish between benign and malicious domains.

1) MALICIOUS DOMAIN GENERATION USING CHARBOT

We referenced the Charbot model by Peck et al. [29] to transform a benign domain into a malicious domain through a few simple steps and does not take much time. It randomly selects two positions in the second-level domain and swaps each with random characters, after that, it changes the top-level domain by random top-level domain selected from the list defined. After two simple steps, from the original legitimate domain, the model generates a malicious domain similar to a benign domain but in this research, we select randomly one position to replace instead of two to make WGAN catch the structure of the target domain and train with lower epochs.

	a	b	c	d	e	.	-	_
f	1	0	0	0	0	0	0	0
a	1	1	0	0	0	0	0	0
c	1	0	0	1	0	0	0	0
e	1	0	0	0	0	0	0	1
i	1	0	0	0	0	0	0	0
d	1	0	0	0	1	0	0	0
.	1	0	0	0	0	0	1	0
v	1	0	0	0	0	0	0	0
n	1	0	0	0	0	0	0	0
⋮								
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

FIGURE 4. Matrix convert from domain.

The list TLD can contain several general top-level domains such as com, at, uk, pl, be, biz, co, jp, cz, de, eu, fr, info, it, ru, lv, me, name, net, nz, org, us. The possible characters set used to be swapped are in the ranges 'a'-'z', '0'-'9', and character '-'.

2) BLACKBOX DETECTOR AND VALIDATOR

The Blackbox detector is the target detector that the Generator needs to bypass. However, the Generator knows nothing about The Blackbox and learns the way to leave the target detector behind through the feedback from the Discriminator. In the framework attack, the Blackbox detector is the target for a direct adversarial attack. In this research, The Blackbox detector can be a DL, EL, or Multimodal model.

The Validator makes sure domains generated by the Generator conform to the basic structure of a domain. We define the basic structure of a domain as some letter cluster alternating some character '.'. For example, 'xxx.xxx' or 'xxxxx.xxx.xxx' conforms to the basic structure, 'xxxxx' or 'xxxx.xx.' is not.

3) DATA PREPROCESSING

The malicious and benign domains are one-hot encoded before being fed into AWG as training data. This is the data digitization step.

We construct a matrix with a shape of 25×40 , where 25 represents the maximum length of a domain, and 40 represents the maximum length of characters that a domain can contain. This is a list comprising lowercase letters 'a' to 'z', digits '0' to '9', and some special characters such as '.', '-', and '_'. Each row in the matrix is defined as one character in a domain, the columns in this row define what characters are in the position. We use a binary matrix, if this position of a domain is the whitespace, the first column in this row is set 0 opposite is set 1. If be set to 1, the column has the same

Algorithm 1 Training WGAN in AWG Framework With Black-Box Phishing Domain Detector

Input

- Domain training set S includes S_{phishing} and S_{benign}
- The number of epochs n_{epoch}
- Generator G with the learning rate lr_G
- Discriminator D with the learning rate lr_D
- Target black-box BB
- The number of critic iterations n_{critic}
- Gradient penalty coefficient λ

Output

- A pair of well-trained Generator G and Discriminator D

- 1: Initialize the Generator G with lr_G
- 2: Initialize the Discriminator D with lr_D
- 3: **for** $i = 0$ to n_{epoch} **do**:
- 4: **for** $j = 0$ to n_{critic} **do**:
- 5: Sample a batch of benign data d_{benign} from S_{benign}
- 6: Sample a batch of AEs data d_{AEs} generated by G
- 7: Update critic D using RMSprop optimizer:
- 8: $\nabla_{\theta_d} \leftarrow \frac{1}{m} \sum_{i=1}^m (BB(d_{\text{benign}}[i]) - BB(d_{\text{AEs}}[i]))$
- 9: $v_d \leftarrow \beta v_d + (1 - \beta)(\nabla_{\theta_d})^2$ $\triangleright$ as in (1)
- 10: $\theta_d \leftarrow \theta_d - lr_D \frac{\nabla_{\theta_d}}{\sqrt{v_d + \epsilon}}$
- 11: Clip critic parameters θ_d within a fixed range
- 12: **end for**
- 13: Sample a batch of AEs data d_{AEs} generated by G
- 14: Update critic G using RMSprop optimizer:
- 15: $\nabla_{\theta_g} \leftarrow \frac{1}{m} \sum_{i=1}^m -G(d_{\text{AEs}}[i])$
- 16: $v_g \leftarrow \beta v_g + (1 - \beta)(\nabla_{\theta_g})^2$ $\triangleright$ as in (2)
- 17: $\theta_g \leftarrow \theta_g - \eta \frac{\nabla_{\theta_g}}{\sqrt{v_g + \epsilon}}$
- 18: **end for**
- 19: **return** G

character as the character in the domain is set to 1. Figure 4 illustrates the process of converting a domain into a matrix.

4) WASSERSTEIN GENERATIVE ADVERSARIAL NETWORKS

We propose using WGAN instead of standard GAN because of some advances in WGAN. The Wasserstein loss function is used in WGAN to help the training process easier and stabler than standard GAN. Moreover, WGAN can avoid some issues in gradient estimation and generate samples with higher quality and more diversity compared to GAN.

The WGAN training process at each epoch will be divided into two parts: the first part involves training the discriminator (D) n_{critic} times, and the second part involves training the generator (G), the details regarding Algorithm 1 with the mathematical symbols clearly explained in Table 3.

a: DISCRIMINATOR

The Discriminator D is a normal classification model. It takes a matrix as an input and predicts this between legitimate and malicious. Because the attacker does not know the external BB , the Discriminator D is utilized to mimic the BB and

TABLE 3. Explanation of mathematical symbols in WGAN training algorithm.

Symbol	Meaning
S	Domain training set including phishing and benign samples
S_{phishing}	Phishing domain training set
S_{benign}	Benign domain training set
n_{epoch}	Number of epochs for training
G	Generator
lr_G	Learning rate of the Generator
D	Discriminator (or critic)
lr_D	Learning rate of the Discriminator
BB	Black-box phishing domain detector
n_{critic}	Number of critic iterations per epoch
λ	Gradient penalty coefficient
∇_{θ_d}	Gradient of the critic (Discriminator)
v_d	Momentum value of the critic
β	Momentum coefficient in the RMSprop optimizer
ϵ	A small value to prevent division by zero in RMSprop
θ_d	Parameters of the critic (Discriminator)
d_{benign}	Batch of benign data sampled from S_{benign}
d_{AEs}	Batch of adversarial examples (AEs) data generated by G
∇_{θ_g}	Gradient of the Generator
v_g	Momentum value of the Generator
θ_g	Parameters of the Generator

provides gradient information to train the Generator G . The input of the Discriminator D consists of adversarial examples from the Generator G and normal examples. Both of them are labeled by the BB first. Then, these results are used as the training dataset labels for the Discriminator D and ensure that the Discriminator D is similar to the BB [28].

The Discriminator D consists of 4 hidden layers with the number of neurons being 64, 128, 128, and 256, respectively. After each hidden layer, Batch Normalization, LeakyReLU, and Dropout layers are added to stabilize and prevent overfitting. Finally, the data is flattened using the Flatten layer and fed into a hidden layer with 1 neuron, utilizing the Linear activation function. The value of output is between -1 and 1 . During the training phase of D , we employ G to generate a batch of synthetic data while loading another batch of randomly sampled real data from the dataset. We then use a black box BB to label these batches. Afterward, we update D using RMSprop optimization, as depicted in 1. Before updating, we clip the critic parameters θ_d within a fixed range. This entire process is repeated n_{critic} times per epoch.

$$\begin{aligned}
 \nabla_{\theta_d} &\leftarrow \frac{1}{m} \sum_{i=1}^m (BB(d_{\text{benign}}[i]) - BB(d_{\text{AEs}}[i])) \\
 v_d &\leftarrow \beta v_d + (1 - \beta)(\nabla_{\theta_d})^2 \\
 \theta_d &\leftarrow \theta_d - lr_D \frac{\nabla_{\theta_d}}{\sqrt{v_d + \epsilon}}
 \end{aligned} \quad (1)$$

b: GENERATOR

The Generator G task is to transform malicious domains generated by the Charbot model into AE domains. It takes a matrix to convert from a domain and noise as input. Noise is a fixed-length vector, and each element is drawn randomly from a Gaussian distribution in the range $[0;1]$.

The Generator G is equipped with two input layers, one for the array and another for the noise. The input branch of the matrix comprises three hidden layers with the number of neurons being 128, 64, and 32, respectively. Meanwhile, the input branch of the noise consists of three hidden layers. The first layer contains 128,000 neurons, followed by a Reshape layer to match the shape of the matrix. The subsequent two layers consist of 64 and 32 neurons, respectively. The activation LeakyReLU and Batch Normalization are added after each hidden layer to stabilize the training process. We use the Concatenate layer to merge the output of two branches and take them into the output layer. The output layer contains N neurons, where N represents the number of elements in the matrix, and it utilizes the Tanh as the activation function. After n_{critic} iterations of training, D is then utilized to evaluate the batch of synthetic data generated by G . Subsequently, this data is used to update G using RMSprop optimization, as detailed in Formula 2.

$$\begin{aligned}
 \nabla_{\theta_g} &\leftarrow \frac{1}{m} \sum_{i=1}^m -G(d_{\text{AEs}}[i]) \\
 v_g &\leftarrow \beta v_g + (1 - \beta)(\nabla_{\theta_g})^2 \\
 \theta_g &\leftarrow \theta_g - lr_G \frac{\nabla_{\theta_g}}{\sqrt{v_g + \epsilon}}
 \end{aligned} \quad (2)$$

C. TRANSFER-BASED ADVERSARIAL ATTACKS

Transfer-based Adversarial Attacks involve leveraging knowledge from a source model to create AEs that can mislead a different model. This exploits the transferability of adversarial perturbations across models, effectively extending the impact of adversarial attacks [62]. This type of attack is frequently employed by attackers, particularly within the context of BB scenarios. The overview of the transferability of adversarial attacks is depicted in Figure 5.

Notably, the study by [63] underscores the effectiveness of targeting ML systems within BB scenarios. Their research demonstrates that if an AE is crafted to deceive model M_1 , it frequently also misleads model M_2 . This inherent transferability property offers a strategic avenue for attacking target models through a deceptive assault on a source model. Researchers have harnessed this inherent property of AEs to implement BB attacks. In one such approach, Papernot et al. [45] construct an attack by training a substitute model to replicate the behavior of the target BB model. Their result shows the victim model is attacked by AEs generated from the substitute model. Additionally, with various ML models such as SVMs, logistic regression models, and neural networks, researchers demonstrated in [45] that AEs successfully transfer, despite the significant differences in their model structures. Moreover, in the research conducted by [28] and his collaborators, the authors incorporate a black box ML-based detector into the GAN training process to produce adversarial URL samples. Their findings reveal that the novel patterns generated by PWDGAN can effectively evade or diminish the effectiveness of ML-based detectors.

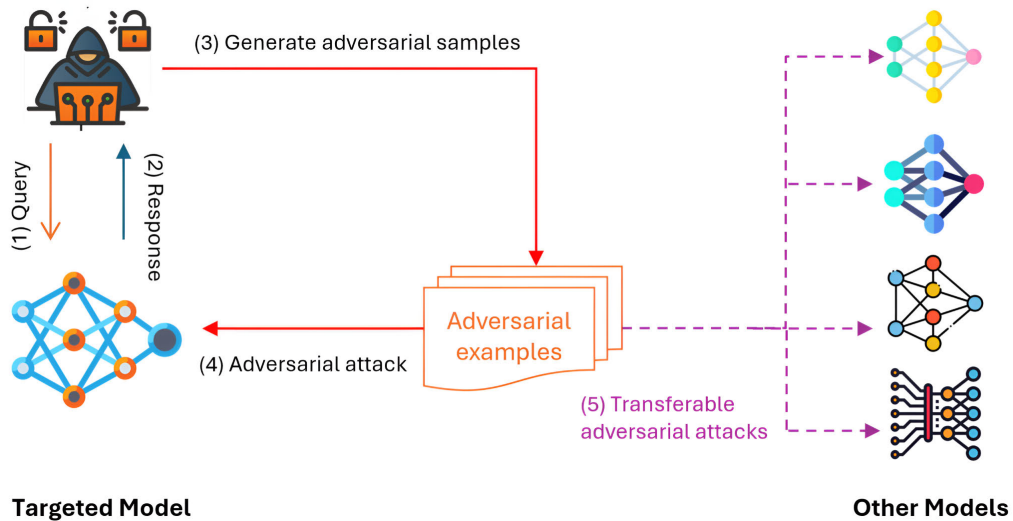


FIGURE 5. The overview of transferring adversarial attacks from a targeted model into other learning models.

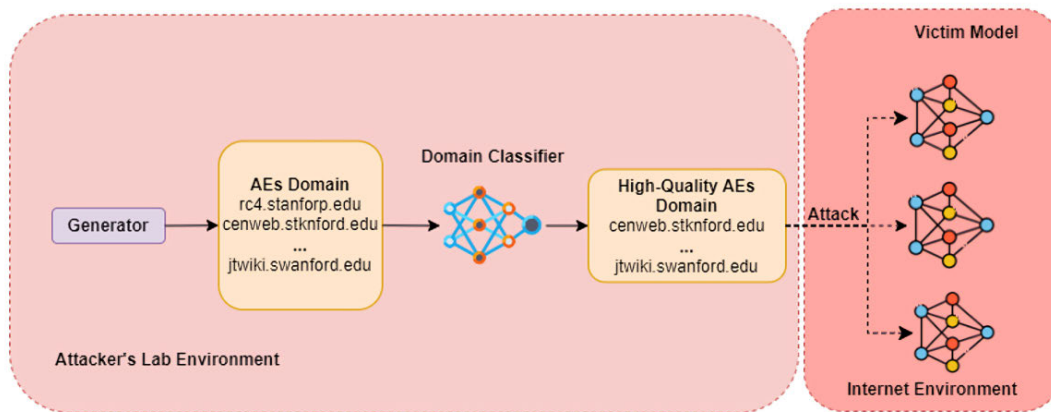


FIGURE 6. The overall of attack phase.

Drawing inspiration from their work, we also employ a WGAN and learning-based black box detector to generate adversarial domains. The generated domains can be utilized to launch attacks against different models. The statistical analysis will be compiled by the team, followed by a comprehensive assessment of the potential of this approach.

In this study, we simulate a phishing attack environment in reality. Our attack method includes the following steps:

- **Step 1: AWG Model Training**

This step involves training the Generator G using our proposed architecture outlined in Figure 3. The outcome of this step will be a Generator G capable of producing highly evasive AEs.

- **Step 2: Selecting adversarial sample instances.**

As depicted in Figure 6, From the attacker's laboratory environment, The domain classifier will test the adversarial example domains generated by the Generator

G . We chose the domain classifier as our testing model for several reasons. Firstly, domain classifiers are typically easy to deploy, lightweight, and highly effective. Secondly, the abundant availability of domain data enhances the prediction accuracy of the models. Hence, the instances bypassed by the domain classifier are considered high-quality samples.

- **Step 3: Launch attacks.**

The high-quality adversarial samples from Step 2 are leveraged to launch attacks on different models on the Internet. These models come from different types of AI, including ML, EL, DL, and MM models. The specifics are illustrated in Figure 6.

D. DEFENSE MECHANISM

Adversarial attack and defense technologies have rapidly evolved, with researchers proposing numerous advanced

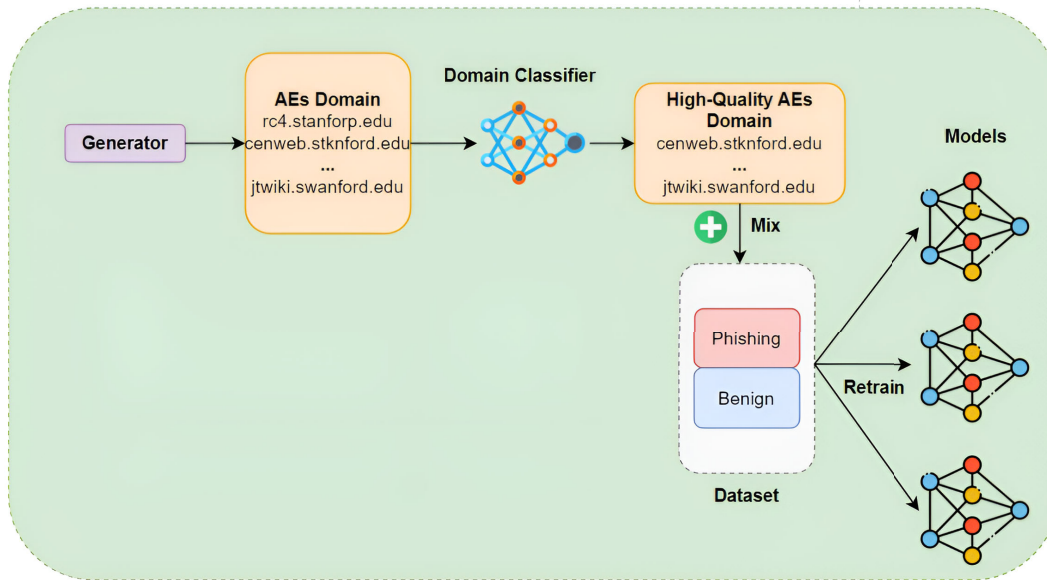


FIGURE 7. The overall of defense phase.

algorithms. One of these is adversarial training [47], [57] [64] which adds AEs into training data to improve model resistance. For instance, Madry et al. [57] utilized examples generated from PGD attacks to augment the training dataset, which remains effective. On the other hand, this method might backfire if excessive retraining is performed with numerous samples, leading to performance degradation. Hence, the adversarial samples need to be carefully selected before being incorporated into the retraining process. Khoda et al. [65] put forward two novel approaches for the selection of malicious samples: one based on the distance from the cluster center, and the other involving probabilities derived from kernel-based learning (KBL). The results of his experiments demonstrate that these approaches have notably assisted the model in accurately distinguishing AEs. Notwithstanding its effectiveness, this approach remains apprehensive due to its computational intensity and the substantial nature of the datasets, as highlighted in [66].

We utilize a simple and user-friendly approach, the Data Optimization technique [55], as the primary defense mechanism in this research. Our defense phase is depicted in Figure 7. To boost the resistance of these models, we employ a retraining strategy with AEs. The high-quality AEs that successfully bypass the domain classifier are mixed into the dataset and utilized to retrain the victim models. This mechanism mitigates data explosion and achieves high efficiency.

V. EXPERIMENTS AND ANALYSIS

This paper is implemented on an Intel(R) Xeon(R) CPU E5-2660 v4 with 32GB RAM using an Ubuntu OS and utilizes Keras with TensorFlow.

TABLE 4. Statistical analysis of domain dataset.

	Source Domain	Number of Domain	Total
Benign	TOP 100 Alexa domain	65,000	110,423
	TOP 10 University domain	44,667	
	TOP 10 Banking domain	756	
Phishing	Phishing Database domain	65,000	110,000
	TOP 10 University domain	45,000	

TABLE 5. Statistical analysis of website dataset.

Type of Website	Number of Website
Phishing	14,201
Benign	9,759
Total	23,960

A. EXPERIMENTAL DATA

We construct two datasets: one for training and testing a phishing domain detector and another for training and testing a phishing website detector. In reality, the attacker lacks knowledge about the training data used for the website classifier on the internet. Thus, the training data for the attacker's domain classifier is distinct from that of the website classifier.

1) DOMAIN DATASET

The first dataset includes benign and phishing domains. Additionally, to collect all the subdomains of a domain, we utilize the Subfinder tool.

The benign dataset is collected from the top 100 Alexa domains, the top 100 University domains, and the top 10 Banking domains in the world. Then, the Subfinder tool is leveraged to get all the subdomains, and we have a total of 110,423 legit domains.

TABLE 6. The models utilized in these experiments.

Target	Type	Model
Domain	Machine Learning	Decision Tree (DT), Support Vector Machine (SVM), and Logistic Regression (LR).
	Ensemble Learning	Random Forest (RF), Adaptive Boosting (AdaBoost), eXtreme Gradient Boosting (XGBoost).
	Deep Learning	Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), CNN-LSTM.
	Mutlimodal	CNN-RNN [27].
Website	Deep Learning	Fusion-CNN [14], CCBLA [68], EMAN-CNN [18]
	Mutlimodal	Shark-Eyes [26], Multi-Modal DL [25]

Next, we collect phishing domains from the Phishing Database repository, updated hourly on GitHub. This is the list of DGA domains generated by 25 different DGA families [67]. Besides, we utilize subdomains from the TOP 10 benign universities and subsequently employ the CharBot [29] algorithm to transform them into phishing subdomains. In total, we gathered 110,000 phishing domains.

Detailed statistics on the number of benign domains and phishing domains are presented in Table 4.

2) WEBSITE DATASET

This dataset comprises both Benign and Phishing websites. We collect phishing URLs from OpenPhish and PhishTank websites, and benign URLs from Alexa. After obtaining the list of URLs, we proceed to request and gather resources from these web pages such as domain, HTML tags, content, and JavaScript. The dataset will include URLs, domains, HTML tags, content, and JavaScript of both Phishing and Benign websites. From Table 5, there are total of 23,960 instances, with 9759 Benign and 14,201 Phishing websites. Collecting diverse resources will enrich the process of reconstructing phishing website detection models, serving well for the development of current multimodal detection models.

B. EVALUATION METRICS

We employ standards metrics including Accuracy, Precision, Recall, and F1-score to assess the performance of models. Furthermore, we utilize the following metrics to evaluate both the attack and defense phases.

1) PERFORMANCE OF GENERATOR

To evaluate the performance of the Generator G , we employ a method of calculating the mean. Each time the Generator G generates m samples, it repeats this process n times. For each generation, we count the number of valid samples x , then compute the ratio $\frac{x}{m}$. Finally, we calculate the average mean after n iterations and get the ratio of valid samples generated by the Generator G . The formula 3 shows the detail.

$$PG = \frac{\sum_{i=1}^n \frac{x}{m}}{n} \quad (3)$$

TABLE 7. Hyperparameters of our models.

Model	Hyperparameters
DT	max_depth: None, min_samples_split: 2, min_samples_leaf: 1, max_features: None
SVM	C: 1.0, kernel: rbf, gamma: scale
LR	C: 1.0, penalty: l2, solver: lbfgs
RF	n_estimators: 100, max_depth: None, min_samples_split: 2, min_samples_leaf: 1, max_features: auto
AdaBoost	base_estimator: DTClassifier(max_depth=1), n_estimators: 50, learning_rate: 1.0
XGBoost	n_estimators: 100, learning_rate: 0.3
CNN	Filters=32, Kernel=(3,3), Activation=relu, Pool=(2,2), Neurons=128, Adam(lr=0.001), Loss=Cross-entropy, Dropout(rate=0.25)
LSTM	Units=default, Activation=sigmoid (gates), tanh (states), Adam(lr=0.001), Loss=Cross-entropy/MSE, Dropout(rate=0.25)
CNN-LSTM	Combined CNN and LSTM architecture

TABLE 8. Hyperparameter of the GAN model.

Hyperparameter	Generator	Discriminator
Optimizer	RMSprop	RMSprop
Activation Function	Tanh	Linear
Learning Rate	0.0001	0.0001
Loss Function	wasserstein_loss	wasserstein_loss
Epochs	5000	5000

2) DETECTION RATES

The detection rates (DR) demonstrate the resilience capability of the models against AEs. This metric is calculated by dividing the total number of detected samples by the total number of input AEs. The Formula expresses the function 4:

$$DR = \frac{n_d}{n} \quad (4)$$

where n is the total number of AEs and n_d is the total number of detected samples.

3) EVASION RATES

The evasion rate (ER) represents the proportion of adversarial samples successfully evading learning-based detectors. The function is defined by the Formula 5:

$$ER = 1 - DR \quad (5)$$

C. EXPERIMENTAL SETTINGS

1) VICTIM MODEL

We built two groups of models: domain classifier and website classifier. Both groups of models will be subjected to assess their ability to resist adversarial attacks.

We employed 4 types of AI technologies including ML, EL, DL, and MM. These models are presented in the Table 6.

For the domain classifier, the hyperparameter information is described in Table 7. We also constructed a multimodal model called CNN-RNN, based on the research [27].

For the website classifier, we rebuilt models from previous studies [14], [18], [25], [68]. These are all state-of-the-art

TABLE 9. The performance of phishing domain detection models.

Target	Type	Models	Accuracy	Precision	Recall	F1-score
Domain	Machine Learning	LR	0.8900	0.9189	0.8720	0.8949
		DT	0.9496	0.9513	0.9496	0.9516
		SVM	0.8984	0.9560	0.8355	0.8916
	Ensemble Learning	RF	0.9656	0.9782	0.9565	0.9674
		AdaBoost	0.912	0.9605	0.8596	0.9072
		XGBoost	0.9249	0.9752	0.8721	0.9208
		CNN	0.9950	0.9995	0.9906	0.9950
	Deep Learning	LSTM	0.9974	0.9972	0.9975	0.9978
		LSTM-CNN	0.9828	0.9907	0.9746	0.9826
	Multimodal	CNN-RNN [27]	0.9943	0.9987	0.9898	0.9942

models with the best current performance. Additionally, we leverage the Shark-Eyes model [26].

2) BLACK-BOX MODEL

To facilitate the transfer attack, we selected two models as *BB*. models based on their performance. As evident from Table 9, the RF model exhibited the best performance within the EL group, while the MM outperformed the others. Hence, we choose RF and MM as a *BB*. model in our experiment.

3) AWG-BASED ADVERSARIAL SAMPLES GENERATION

We employed AWG with the architecture details of the Generator *G*. and Discriminator *D*. outlined in Table 8. The model was trained for a total of 5000 epochs, with n_{critic} set to 5 *D*. trained 5 times more than *G*., clamp set to 0.01. The RMSprop function was used as the optimization function in the model, with a learning rate of 0.0001.

D. EXPERIMENT SCENARIOS

We proposed five scenarios for comprehensive evaluation. These experiments aim to evaluate whether our proposed model can be used in real-world applications.

- **Scenario 1:** *The performance of domain classifier.*
 - In this part, we evaluate the ability of models to detect phishing domains. The domain classifier is utilized in the attack phase and assessed for its ability to withstand adversarial attacks.
- **Scenario 2:** *The performance analysis of learning-based models in website phishing detection.*
 - We evaluate the ability of learning-based models to detect phishing websites and explore the balance between their effectiveness and resource consumption. These models serve to simulate victims on the Internet.
- **Scenario 3:** *The transfer adversarial attack analysis.*
 - In this experiment, we utilize the Generator *G* trained with the MM (GAN MM) models and RF models (GAN RF) as the black-box model *BB*. participates in the training process. We established an ensemble model using the voting method, with selected member models including RF, MM, DT, and LSTM. These are the best-performing models in our representative group. Subsequently, the AEs that bypass the ensemble model

will be collected to execute transfer attacks on various models.

- **Scenario 4:** *The performance analysis of AWG model.*
 - In this work, we investigate the performance of the Generator *G* by various aspects. Moreover, we investigate the evasion rate of our AEs when facing a system. There are two tasks in this experiment.

- Task 1: Transfer attack on the domain detector.
- Task 2: Transfer attack on the website detector.

The outcome of this scenario comprises statistical data on the resistance capability against AEs of both the phishing domain and website detector.

- **Scenario 5:** *Boosting Resistance for the Victim model*
 - We provide findings on the resilience of AI-based models after employing our defense mechanism.

E. EXPERIMENT RESULTS:

The results from these scenarios are presented as follows:

1) SCENARIO 1: THE PERFORMANCE ANALYSIS OF LEARNING-BASED MODELS IN DOMAIN PHISHING DETECTION

As seen in Table 9, SVM and LR exhibit comparable detection capabilities within the ML group, with the highest F1-score reaching 89.49%. In the EL group, RF demonstrates the highest ability in detecting phishing domains, with accuracy and F1-score reaching 96.56% and 96.74%, respectively. In contrast, AdaBoost is the least effective, showcasing an accuracy of 91.2% and an F1-score of 90.72%. In the DL group, the models demonstrate superior predictive capabilities, with all metrics surpassing 97%. Among them, CNN stands out with outstanding performance, achieving an accuracy of 99.74% and an F1-score of 99.78%. In the final group, MM stands out as the most effective predictor, outperforming other models in all metrics. More precisely, MM attains scores of 99.43%, 99.87%, 98.98%, and 99.42% for accuracy, precision, recall, and F1-score, respectively.

The results of this scenario showcase that all proposed models excel in identifying phishing domains. These models are constructed to serve the adversary's attack phase. Therefore, the better their performance, the higher the effectiveness of the attack phase.

TABLE 10. The performance of phishing website detection models.

Target	Type	Model	Accuracy	Recall	Precision	F1-score
URL	Deep Learning	CCBLA [68]	0.9266	0.9303	0.9348	0.9325
		Eman-CNN [18]	0.9348	0.9739	0.9144	0.9432
		Fusion-CNN [14]	0.9458	0.9814	0.9256	0.9527
URL, HTML	Multimodal	Shark-Eyes [26]	0.9347	0.9577	0.9303	0.9438
URL, Content, Javascript		Multi-Modal DL [25]	0.9340	0.9459	0.94195	0.9439

TABLE 11. Statistics on the time and resource consumption of the learning-based models.

Model	Type	Numbers Website	Time (sec)	RAM Used (MB)
CCBLA [68]	Deep Learning	1000	15,9061	106,9804
		2000	23,9169	157,4343
Fusion-CNN [14]		1000	1,0934	9,425
		2000	1,9257	17,539
Eman-CNN [18]		1000	10,5535	55,70625
		2000	20,47169	93,88906
Multi-modal DL [25]	Multimodal	1000	238,04804	13,5914
		2000	380,8768	23,10538
Shark-Eyes [26]		1000	144,6243	120,9296
		2000	296,1614	143,0882

2) SCENARIO 2: THE PERFORMANCE ANALYSIS OF LEARNING-BASED MODELS IN WEBSITE PHISHING DETECTION

The results in Table 10 indicate that all models achieve high effectiveness in detecting phishing websites, with accuracy above 92%. Among them, in the group of models targeting URLs, the Fusion-CNN model [14] achieves the highest F1-score of 95.27%, while the CCBLA model [68] achieves the lowest F1-score of 93.25%. Within the realm of Multimodal Models, accuracy surpassing 93.4% has been achieved. For instance, the Shark-Eyes model [26] focusing on URLs and HTML, also demonstrates strong performance with an F1-score of 94.38%. Additionally, the Multi-Modal DL model [25], targeting URL, content, and JavaScript components, also achieves a relatively high F1-score of 94.39%.

In addition, runtime and memory consumption are factors we aim to investigate. We conducted experiments with input sizes of 1000 websites and 2000 websites to measure the time taken for predictions and the RAM used by the models. From Table 11, overall, unimodal DL models show lower prediction times and memory consumption compared to MM models. This can be explained by the fact that MM models handle multiple input branches, making the processing more complex and resource-intensive. Among DL models, the Fusion-CNN [14] model exhibits the lowest time and RAM consumption, with only 1.0934 seconds and 9.425 MB RAM used for 1000 websites. Conversely, in the MM category, Shark-Eyes [26] runs faster than Multi-modal DL [25] but consumes more RAM. For instance, when given 1000 websites as input, Shark-Eyes [26] takes approximately 145 seconds, whereas Multi-modal DL [25] takes about 238 seconds. However, Shark-Eyes [26] consumes approximately 121 MB of RAM compared to only 23 MB for Multi-modal DL [25].

The findings underscore the high effectiveness of our approach in detecting phishing websites. These models

TABLE 12. The detection rates of domain classifier.

Type	Models	GAN MM	GAN RF
Machine Learning	LR	0.6016	0.9907
	DT	0.5111	0.9702
	SVM	0.1367	0.1002
Ensemble Learning	RF	0.5312	0.9814
	AdaBoost	0.3653	0.8560
	XGBoost	0.1924	0.3832
Deep Learning	CNN	0.9889	0.9812
	LSTM	0.9893	0.9879
	LSTM-CNN	0.7389	0.8750
Multimodal	CNN-RNN [27]	0.9782	0.9716

TABLE 13. The detection rates of website classifier.

Models	GAN MM			GAN RF		
	100	1000	5000	100	1000	5000
Shark-Eyes [26]	0.99	0.972	0.973	0.98	0.967	0.976
Multi-Modal DL [25]	0.97	0.955	0.960	0.95	0.952	0.961
CCBLA [68]	0.39	0.497	0.5242	0.86	0.884	0.879
Eman-CNN [18]	0.13	0.12	0.1348	0.3	0.297	0.293
Fusion-CNN [14]	0.3	0.151	0.138	0.55	0.332	0.292

simulate AI-based phishing detection systems, achieving optimal performance that accurately reflects real-world scenarios. Furthermore, it is important to note that MM models a trade-off between effectiveness and resource/time consumption. The increased resource consumption in exchange for better prediction capabilities against AEs is promising for MMs. Moreover, with real-world infrastructure, the consumption levels of MLs are not excessively high and can be adequately managed. Therefore, future research has significant potential to improve multimodal architectures to reduce resource consumption.

3) SCENARIO 3: THE TRANSFER-BASED ADVERSARIAL ATTACK ANALYSIS

The experimental results are detailed in Table 12 in the first case. Specifically, the AEs of GAN MM demonstrate effective evasion across all 2 groups: EL, and ML. In the ML group, the SVM model exhibits the lowest defense

TABLE 14. The statistics on the rate of producing valid samples of G .

No. AEs / 1 generation	No. Iterations	Rate of Valid Sample
100	100	90.1 %
100	500	90.9 %
100	1000	90.02 %
100	5000	90 %

capability, achieving a detection rate of only 13.67%, whereas LR achieves the highest detection rate at 60.16%. Within the EL group, the RF model demonstrates a detection accuracy of 53.12% against adversarial patterns, while XGBoost performs the least effectively, achieving only 19.24%. Conversely, the DL and Multimodal groups show-case robust resilience against GAN MM. However, within the DL group, the LSTM-CNN model is still vulnerable to some AEs, resulting in a reduced detection rate of 73.89%. On the other hand, the multimodal CNN-RNN model [27] exhibits superior resilience, with a detection rate of 97.82%.

In the second one, studies have revealed that fraudulent websites are often identified based on URLs, a method that is not only lightweight but also highly effective. However, the URL-based detection method has not yet been validated for its resilience against adversarial attacks. Therefore, we conducted a study to research and rebuild the fraudulent website classification models based on previous research.

We sequentially tested on sets of 100, 1000, and 5000 domain AEs to obtain comprehensive results. The details are described in Table 13. Overall, the Shark-Eyes model [26] demonstrates better capabilities in detecting AEs than others. When faced with AEs generated by GAN MM, the multimodal Shark-Eyes model [26] demonstrates the highest accuracy, achieving a detection rate of 99% for 100 input domains and the lowest of 97.2% for 1000 AEs. Furthermore, the Multi-Modal DL [25] also exhibits strong performance in identifying adversarial samples, achieving an average detection rate of 96%. Conversely, models targeting only the URL exhibit poor defense against AEs. Specifically, the Eman-CNN model [18] shows the lowest detection rates at 13%, 12%, and 13.48% for 100, 1000, and 5000 inputs, respectively. Even the Fusion-CNN model, which has the highest F1-Score according to Table 10, suffers from AE attacks, resulting in a detection rate of only 12% for 1000 input AEs. Moreover, regarding AEs generated by GAN RF, the CCBLA model showcased the highest detection capability at 88.4%, whereas the Eman-CNN [18] and Fusion-CNN [14] models achieved 30% and 55%, respectively. The Shark-Eyes model [26] continues to outperform, with detection rates all above 96.7%. Moreover, the Multi-Modal DL [25] also effectively handles AEs from the GAN RF, boasting a peak detection rate of 96.1% when presented with 5000 input AEs. Due to these Multimodal models focusing on multiple aspects of the input. Thus, it can effectively thwart scenarios where attackers craft AEs targeting specific inputs, such as URLs.

TABLE 15. Examples of AEs generated by AWG trained with MM and RF models as BB.

GAN MM	GAN RF
ks-dev.rm.f.harvaid.edu	darseng.ntandford.edu
cedback.lib.harvard.0du	dord25.sganford.sdu
sumni.harbard.edu	sop.stanford.tde
prkes.seas.haqard.edu	lvtwet.sta0ord.ddu

TABLE 16. Statistics on the time and resources consumed during the AE generation process.

Time (second)	Number of generated AEs	Memory used (MB)
1	30	0,9216
30	747	3,7888
60	1475	4,7104

TABLE 17. The evasion rate of AEs against the phishing detector.

	Model	Evasion Rate
Domain Classifier	LR	0.3984
	DT	0.4889
	SVM	0.8633
	RF	0.4688
	AdaBoost	0.6347
	XGBoost	0.8076
	CNN	0.0111
	LSTM	0.0107
	CNN-LSTM	0.2611
Website Classifier	CNN-RNN [27]	0.0218
	CCBLA [68]	0.503
	Fusion-CNN [14]	0.849
	Eman-CNN [18]	0.88
	Multi-Modal DL [25]	0.045
	Shark-Eyes [26]	0.028

From two tasks, although all ML, EL, and DL models are susceptible to noise from AEs, the MMs maintain a high capability to detect AEs.

Nonetheless, all models are affected by AEs. In phishing attacks, the risk is substantial as a single domain evading detectors could lead to a successful attack. Therefore, it is essential to develop a strategy to counter AEs effectively.

4) SCENARIO 4: THE PERFORMANCE ANALYSIS OF THE AWG MODEL

The Generator G after training can generate domain names based on input domains. We evaluate the performance of the Generator G based on the following criteria:

- The rate of generating valid samples.
- The similarity of the generated AEs to authentic ones.
- The time and resources consumed during the AE generation process.

First, we choose “stanford.edu” and “harvard.edu” as the base input for G . In reality, any domain name can be selected. Next, we calculate the ratio of valid AEs by creating 100 samples and repeating from 100 to 5000 times for this process before averaging the results. As shown in Table 14, the results indicate that the capability of generating valid AEs by G surpasses 90%. Furthermore, we provide a set

TABLE 18. The performance of models after applying our defense mechanism.

Function	Model	Accuracy	Precision	Recall	F1_score	Detection Rate
AEs Domain Detection	RF	0.9613	0.9647	0.9440	0.9538	0.9868
	DT	0.9298	0.9494	0.9381	0.9425	0.9613
	LSTM-CNN	0.9794	0.9855	0.9814	0.9811	0.9097
	CNN-RNN [27]	0.9803	0.9880	0.9787	0.9792	0.9873
AEs Website Detection	CCBLA [68]	0.8787	0.9021	0.9036	0.9028	0.9166
	Fusion-CNN [14]	0.9503	0.9608	0.9594	0.9601	0.9997
	Eman-CNN [18]	0.9319	0.9500	0.9393	0.9446	0.9987
	Multi-modal DL [25]	0.9301	0.9383	0.9529	0.9355	0.9767
	Shark-Eyes [26]	0.9410	0.9546	0.9447	0.9497	0.9978

of domains generated from G trained with GAN MM and GAN RF in Table 15. It can be observed that the AEs exhibit a valid structure and closely resemble their original domains. Lastly, the statistics on resource consumption are presented in Table 16. We gradually increased the runtime from 1 second to 60 seconds to analyze the number of AEs created during that time and the amount of memory they consumed. The results indicate that as the runtime increases, the number of generated samples also increases, accompanied by a rise in memory usage. Specifically, with a runtime of 1 minute, the Generator G produced 1475 AEs and consumed 4.7104 MB of memory. This memory usage is deemed acceptable considering the effectiveness of the G .

Additionally, we reference Table 12 and Table 13 to compute the evasion rate of our AEs. We use the domain classifier's detection rate against GAN MM and the web classifier's detection rate against 1000 GAN MM inputs. From Table 17, both classifier's performance is affected, with our AEs exhibiting the highest evasion rate: 86.33% against the SVM model in the domain detector group and 88% against the Eman-CNN model [18] in the website detector group.

These findings highlight the effectiveness of our Generator, as evidenced not only by its evasive rate but also by its capability to generate up to 1475 samples per minute, demonstrating its efficient data enrichment capability.

5) SCENARIO 5: BOOSTING AES RESISTANCE FOR THE VICTIM MODEL

Specifically, we generated 10,000 new AEs, consisting of 5,000 AEs from GAN MM and 5,000 AEs from GAN RF. These samples were merged into the original training set and used for retraining the models. In the domain adversarial detection function, we opted to retrain RF, DT, CNN-LSTM, and CNN-RNN models [27]. Subsequently, for the remaining functions, all four MMs were retrained. We create new 100,000 AEs from both GAN MM and GAN RF. These AEs will be used to validate the resistance of the Victim models.

The results presented in Table 18 illustrate the enhanced resilience of these models when confronted with AEs. In particular, the detection rates of these models are all above 90.97%. Among them, the Eman-CNN model [18] shows the most significant improvement, increasing from 13% to 99.87%. Additionally, the CNN-LSTM model also

demonstrates some enhancement, rising from 73.89% to 90.97%. The remaining models all exhibit high detection capabilities against AEs after applying our technique. The Fusion-CNN model [14] achieves an exceptional detection rate, approaching 99.97%. We also included four common metrics—accuracy, precision, recall, and F1-score—to evaluate the models after applying our defense mechanism. In the AEs domain detection group, the CNN-RNN model [27] achieved the highest detection rate of 98.73%, with the lowest of the other four metrics being 97.87% recall. For the AEs website detection group, the Fusion-CNN model [14] achieved the highest detection rate of 98.73%, with the lowest of the other four metrics being 95.03% accuracy. These results demonstrate the effectiveness of our defense strategy, which not only enhances the detection rates but also ensures that other metrics remain high.

VI. DISCUSSION

A. ETHICAL AND PRIVACY ISSUES

The paper addresses the privacy and ethical challenges associated with the creation and application of adversarial samples for phishing website detection using URL and HTML tags in ML/DL models. It underscores the critical importance of data stewardship by enforcing rigorous data protection measures aligned with prevailing privacy laws and best practices. Since crawling from public websites without login activities, our work includes the anonymization of personally identifiable information (PII) to ensure anonymity and prevent the traceability of data back to individual users, thus preserving user privacy. Moreover, the document provides detailed insights into the generation and management of adversarial datasets. These datasets are either synthetically produced or derived from modifying existing data, utilizing techniques that subtly alter inputs such as URLs and HTML tags. Critically, during the generation of these datasets, care is taken to strip any sensitive data, thereby upholding the confidentiality of the information and ensuring full compliance with data protection regulations. This approach not only bolsters the integrity and robustness of phishing detection systems but also maintains a firm commitment to ethical practices and the safeguarding of user data throughout the research and application phases.

Our paper also emphasizes the use of adversarial samples strictly within controlled environments to evaluate and enhance the robustness of phishing detection models without

TABLE 19. Discuss limitation and future direction.

Criteria	Weakness	Future Research Direction
Diversity	Dataset: Due to the typically short lifespan of phishing websites, the number of collected phishing websites is still limited.	Develop a bot that continuously updates our dataset with phishing websites reported on reputable platforms, allowing us to retrain our models regularly. Contribute our collected data to the research community for future work.
	AEs generated from AWG: There might exist more sophisticated and sophisticated techniques that our AWG framework may not cover, such as homograph attacks.	Enhance the architecture of proposed AWG and integrate advanced attack techniques such as homograph attack and utilizing Explainable Artificial Intelligence to achieve higher diversity and realism, making our models more robust and applicable to real-world scenarios.
Effectiveness	Multimodal models: In this study, the URL is an AEs, but the HTML tags are not manipulated so the model can still make accurate decisions by utilizing the normal branch. However, there is a lack of scenarios where all input are adversarial. Next, our proposed MMs uses both URL and HTML data to make predictions, calculating the final output based on predictions from both sources. If the URL dataset is of poor quality, even a good HTML dataset cannot prevent the calculation process from being affected, leading to false positives or false negatives.	Conduct another study on a different attack case by applying the Multimodal GAN model [69] to generate both adversarial URL and HTML tags. Use our AWG attack framework to generate samples with diverse patterns, enhance the dataset, and stabilize the MMs.
	AWG Attack Framework: The ability to bypass DL and MM models remains limited.	Enhance the AWG architecture by utilizing MultiGAN [69] or StyleGAN3 and fine-tuning the model parameters.
	Defense mechanism: Although our proposed method is simple, and effective, other exploitation techniques may become more sophisticated and complex in the future, potentially bypassing our mechanism.	New defense strategies or advancements to our defense mechanism will be required for better protection. Develop an automated pipeline encompassing attack framework, multimodal, and defense phases. The goal is to automate the data generation process and utilize defense mechanisms to ensure our models are always resilient against advanced attacks
Long-term sustainability and Maintenance	Our defense mechanism can adapt to new threats by retraining with phishing samples from around the world and AEs generated by the AWG Framework. Our attack phase generates new AEs to help learning-based models continually adapt to emerging threats. However, the automatic sample generation pipeline for the AWG Framework and its integration into the defense phase is not yet complete.	Integrate AWG Attack Framework and Defense mechanism into an automated pipeline.
Compatibility and Interoperability	MMs are designed to be compact, portable, and highly accurate, making them easy to integrate into existing systems. However, there is still a trade-off between MMs' performance and runtime/memory usage.	Integrate models into browser extensions such as Firefox, Chrome, and Microsoft Edge. Propose an interface or API for seamless integration of our models into existing phishing detection systems.
Complexity	Building and training our attack frameworks and effective defense strategies requires practitioners to have in-depth knowledge of ML and cybersecurity, a sufficiently powerful system, and a comprehensive dataset for training and testing. Because we are using state-of-the-art models, building our models may not be feasible for beginners.	Write a document to help practitioner use our frameworks step by step.

compromising the integrity of actual user account data. It is crucial to be aware of the potential misuse of adversarial techniques, and advocate for the development of ethical guidelines and standards to govern the use of such methods in security research. Some of the overall guidelines can be conducted as follows:

- **Purpose Limitation:** Adversarial techniques should only be employed for legitimate and ethical purposes, such as improving the robustness and efficacy of phishing detection models. They should not be used to exploit vulnerabilities for malicious purposes or to deceive legitimate users or entities.

- **Consent and Anonymity:** Whenever real user data is involved, researchers should obtain consent from participants for the use of their data in generating adversarial samples. Furthermore, data should be anonymized and de-identified to prevent any potential privacy breaches and to ensure compliance with privacy laws and regulations.
- **Risk Assessment and Mitigation:** Prior to the deployment of adversarial samples, a thorough risk assessment should be conducted to evaluate potential impacts on the security systems and the users they protect. Mitigation strategies should be developed to address identified risks, such as unintended model behaviors or increased false positives/negatives in phishing detection.
- **Compliance with Ethical Standards:** The development and use of adversarial samples should comply with ethical standards and guidelines established by relevant professional organizations and oversight bodies. These standards serve as a baseline for conducting research responsibly and ethically.
- **Education and Awareness:** Promote education and awareness about the ethical use of adversarial techniques among researchers, practitioners, and stakeholders. Training programs and workshops can help stakeholders understand the ethical dimensions and implications of using adversarial samples in research.
- **Collaboration:** Encourage collaboration between industry, academia, and regulatory bodies to continuously refine and update the ethical guidelines as new challenges and technological advances emerge.

By taking these steps, the research aims to leverage the benefits of adversarial samples for improving detection mechanisms while minimizing the risk of harm and maintaining a strong commitment to ethical principles and user privacy.

B. LIMITATION AND FUTURE DIRECTION

In this section, we discuss the limitations of our research and outline potential directions for future studies. We believe that identifying our weaknesses and suggesting future research directions will illuminate the path for future studies. The detailed aspect of the characteristics of diversity, effectiveness, long-term sustainability, maintenance, compatibility, interoperability, and complexity of our work is provided in Table 19.

VII. CONCLUSION

In this study, we introduce an AWG model for generating adversarial samples by leveraging the concept of transfer attacks against phishing website detection models. These samples not only replicate the original structure but also exhibit high evasion capabilities. Specifically, the generated AEs, in conjunction with our attack phase, can effectively target most state-of-the-art PWD models, including ML, DL, EL and MM. This poses a significant challenge for

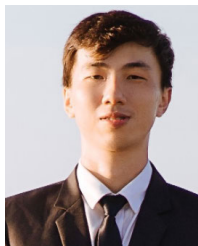
scam detection systems based on blocklist datasets or AI models. Furthermore, AEs generated from our attack phase can serve as valuable resources for training and helping AI models adapt to new cyber threats. Besides, MM demonstrate superior resistance to adversarial attacks, with the highest detection rate compared to others. Lastly, our defense strategy is acknowledged as a straightforward, easy-to-implement, highly effective approach. Nonetheless, one adversarial website bypassing the detector in phishing attacks can have significant consequences. Therefore, in the future, we aim to refine our attack and defense methods further to bolster the robustness of AI models. Ultimately, our study seeks to pioneer advancements on building more robust learning-based PWD in the realm of adversarial phishing detection research.

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PHAN THE DUY received the B.Eng. and M.Sc. degrees in software engineering and information technology from the University of Information Technology, Vietnam National University Ho Chi Minh City (UIT-VNU-HCM), Hochiminh City, Vietnam, in 2013 and 2016, respectively. Currently, he is pursuing the Ph.D. degree in information technology, specializing in cybersecurity with UIT-VNU-HCM. Since 2016, he has been a Research Member with the Information Security Laboratory (InSecLab), UIT-VNU-HCM after five years in the industry, where he joined and created several security-enhanced and large-scale teleconference systems. His main research interests concentrate on cybersecurity and privacy problems, including software security, software-defined networking (SDN), malware and cyber threat detection, digital forensics, machine learning and adversarial machine learning in cybersecurity domain, private machine learning, and blockchain.



VO QUANG MINH is currently in information security program with the University of Information Technology, Vietnam National University Ho Chi Minh City (UIT-VNU-HCM), Ho Chi Minh City, Vietnam. Since 2022, he has been actively involved as a Student Member with the Information Security Laboratory (InSecLab), UIT-VNU-HCM, where he dedicates his efforts to research in the field of AI-based security. His primary research interests include adversarial attacks, multimodal approaches, and phishing within the realm of cybersecurity.



approaches, and federated learning in cybersecurity.

BUI TAN HAI DANG is currently pursuing the degree in information security with the University of Information Technology, Vietnam National University Ho Chi Minh City (UIT-VNU-HCM), Ho Chi Minh City, Vietnam. Since 2022, he has been a Student Member with the Information Security Laboratory (InSecLab), UIT-VNU-HCM, where he focuses on conducting research in the AI-based security domain. His primary research interests include adversarial attacks, multimodal



NGO DUC HOANG SON is currently pursuing the degree specializing in information security with the University of Information Technology, Vietnam National University Ho Chi Minh City, Vietnam (UIT-VNU-HCM), Ho Chi Minh City, Vietnam. Since 2022, he has been actively engaged as a Student Member with the Information Security Laboratory (InSecLab), UIT-VNU-HCM, focusing on projects related to information security and AI-driven security. His primary research interests include AI cybersecurity endeavors, particularly in the realms of malware detection and software-defined networking (SDN).



NGUYEN HUU QUYEN is currently pursuing the degree in computer science with the University of Information Technology, Vietnam National University Ho Chi Minh City (UIT-VNU-HCM), Ho Chi Minh City, Vietnam. Since 2021, he has been a Student Member in the tasks of information security and AI-based security with the Information Security Laboratory (InSecLab), UIT-VNU-HCM. His main research interests include adversarial attacks on machine learning algorithms and federated Learning.



VAN-HAU PHAM received the bachelor's degree in computer science from the University of Natural Sciences of Ho Chi Minh City, in 1998, and the master's degree in computer science from Institut de la Francophonie pour l'Informatique (IFI), Vietnam, in 2004. Then, he did his Internship and was a full-time Research Engineer in France for two years. He then persuaded the Ph.D. thesis on network security under the direction of Prof. Marc Dacier, from 2005 to 2009. He is currently a Lecturer with the University of Information Technology, Vietnam National University Ho Chi Minh City (UIT-VNU-HCM), Ho Chi Minh City, Vietnam. His main research interests include network security, system security, mobile security, and cloud computing.

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